HOMEWORK 4

1) **Consider the LSI system with unit sample response**
   $h[n]=(1/2)^{n-1}$ (u[n-1] - u[n-n_0])   where $4< n_0 < \infty$.
   a) **Is this system stable? Prove or disprove.**
   b) **Is this system causal? Prove or disprove.**
   c) **Compute the outputs of this system to the inputs below and sketch y[n] (you may assume n_0 =5 for the sketches)**
      i) **x[n]= p_4[n]**   ii) **x[n]= $(1/3)^{n-1}$ u[n]**
2) **Repeat question 1 for h[n] = $(1/2)^{n-1}$ u[n-1] – $(-2)^n$ u[-n]**
3) **Consider the systems with the unit sample responses**
   $h_1[n]$ = ¼ δ[n-1] + ½ δ[n] + ¼ δ[n+1]
   $h_2[n]$ = $h_1[n-1]$ = ¼ δ[n-2] + ½ δ[n-1] + ¼ δ[n]
   $h_3[n]$ = -¼ δ[n-1] + ½ δ[n] - ¼ δ[n+1]

   a) **Sketch $h_1[n]$, $h_2[n]$ and $h_3[n]$ one under the other. Observe the coefficients.**
   b) **Determine the corresponding DTFTs, $H_1(e^{jw})$, $H_2(e^{jw})$ and $H_3(e^{jw})$.**
   c) **Write down the expressions for the magnitude and phase responses.**
   d) **Sketch magnitude and phase spectra of $H_1(e^{jw})$, $H_2(e^{jw})$ and $H_3(e^{jw})$. (You may find the identity $\tan(\pi/2 - \alpha) = \cot(\alpha)$ useful for phase spectrum)**
   e) **Compare and contrast the magnitude and phase responses of the filters. How are they related?**